2026Fall Quiz

for (MATH261) Discrete Mathematics

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Problem 1.1. Let $X = \mathbb{R}$ and let $R = \varnothing$ be the void relation on X. Determine whether R is reflexive, symmetric, and transitive, and hence whether R is an equivalence relation. Justify each answer.

Solution. The relation R is not reflexive, since $(x, x) \notin R$ for every $x \in X$.

The relation R is symmetric: the implication $xRy \Rightarrow yRx$ is vacuously true because xRy never occurs. Likewise, R is transitive because the premise $xRy \wedge yRz$ never occurs, so

$$(xRy \wedge yRz) \Rightarrow xRz$$

is vacuously true.

Thus R is symmetric and transitive but not reflexive. Therefore R is not an equivalence relation.

□

Problem 1.2. For $n \in \mathbb{N}_0$, prove that

$$\sum_{i=0}^{n} \binom{n}{i}^2 = \binom{2n}{n}.$$

Solution 1. Let U and V be disjoint sets with $|U| = |V| = n$. Count the n-element subsets of $U \cup V$ according to the number i of selected elements from U. If exactly i elements are chosen from U, then $n - i$ elements must be chosen from V, giving

$$\binom{n}{i}\binom{n}{n-i} = \binom{n}{i}^2$$

choices. Summing over $i = 0, \ldots, n$ counts all n-element subsets of the $2n$-element set $U \cup V$. Hence

$$\sum_{i=0}^{n} \binom{n}{i}^2 = \binom{2n}{n}.$$

□

Solution 2. By symmetry of binomial coefficients and Vandermonde's identity,

$$\sum_{i=0}^{n} \binom{n}{i}^2 = \sum_{i=0}^{n} \binom{n}{i}\binom{n}{n-i} = \binom{2n}{n}.$$

□

Problem 2.1. A positive integer a is called *square-free* if it is not divisible by any perfect square other than 1. For example, 1, 7, and 30 are square-free, whereas 4 and 63 are not. Count the square-free positive integers less than or equal to 100.

Solution. A positive integer $a \le 100$ is square-free if and only if none of 4, 9, 25, and 49 divides a. If one of these prime squares divides a, then a is not square-free. Conversely, if a is not square-free, then $b^2 \mid a$ for some integer $b \ge 2$. Choose a prime $p \mid b$. Since $p^2 \mid a \le 100$, we have $p \le 10$, so $p \in \{2, 3, 5, 7\}$.

For $p \in \{2, 3, 5, 7\}$, define

$$A_p = \{a \in [100] \mid p^2 \mid a\}.$$

The sizes of these sets are

$$|A_2| = \left\lfloor \frac{100}{4} \right\rfloor = 25, \quad |A_3| = \left\lfloor \frac{100}{9} \right\rfloor = 11, \quad |A_5| = 4, \quad |A_7| = 2.$$

The only nonempty pairwise intersections are

$$|A_2 \cap A_3| = \left\lfloor \frac{100}{36} \right\rfloor = 2, \qquad |A_2 \cap A_5| = \left\lfloor \frac{100}{100} \right\rfloor = 1.$$

All other pairwise intersections, and hence all intersections of three or more sets, are empty: the product of the corresponding distinct prime squares exceeds 100. Therefore, by inclusion-exclusion, the number of square-free positive integers at most 100 is

$$100 - |A_2 \cup A_3 \cup A_5 \cup A_7| = 100 - (25 + 11 + 4 + 2) + (2 + 1) = 61.$$

□

Problem 2.2. Let $n, k \in \mathbb{N}$. Count the sequences $(a_1, \ldots, a_k)$ satisfying

$$\{a_1, \ldots, a_k\} \subseteq [n], \qquad a_1 < \cdots < a_k, \qquad |a_i - a_j| \ge 2 \quad (i \ne j).$$

Solution. The conditions imply $a_{i+1} \ge a_i + 2$. Since $a_1 \ge 1$, it follows that $a_k \ge 2k - 1$. Thus no such sequence exists if $n < 2k - 1$.

Suppose $n \ge 2k - 1$. For each admissible sequence, define

$$b_i = a_i - (i - 1), \qquad 1 \le i \le k.$$

Then

$$1 \le b_1 < b_2 < \cdots < b_k \le n - k + 1.$$

Conversely, given any strictly increasing sequence $1 \le b_1 < \cdots < b_k \le n - k + 1$, the formula $a_i = b_i + (i - 1)$ produces an admissible sequence in $[n]$. These constructions are inverse to each other, so the desired sequences are in bijection with the k-element subsets of $[n - k + 1]$. Hence their number is

$$\begin{cases} \dbinom{n - k + 1}{k}, & n \ge 2k - 1, \\ 0, & n < 2k - 1. \end{cases}$$

□